

# multicomp<sup>PRO</sup>



DC Electronic Load User Manual

Model No. MP710258 and MP710259

## Specifications

Note: The specifications below are tested under the conditions of temperature 25°C±5°C and the warm-up for 20 minutes.

| Models                      | MP710258                    | MP710259                                                    |
|-----------------------------|-----------------------------|-------------------------------------------------------------|
| Input Rating                | Power<br>Voltage<br>Current | 300W<br>0-120V<br>0-30A                                     |
| CC Mode                     | Range                       | 0-3A 0-30A 0-3A 0-30A                                       |
|                             | Resolution                  | 0.1mA 1mA 0.1mA 1mA                                         |
|                             | Accuracy                    | ±(0.05%+0.05% FS) ±(0.05%+0.05% FS)                         |
| CV Mode                     | Range                       | 0-18V 0-120V 0-18V 0-120V                                   |
|                             | Resolution                  | 0.1mV 1mV 0.1mV 1mV                                         |
|                             | Accuracy                    | ±(0.05%+0.025% FS) ±(0.05%+0.025% FS)                       |
| CR Mode                     | Range                       | 0.05Ω-7.5KΩ 0.05Ω-7.5KΩ                                     |
|                             | Resolution                  | 16 bit 16 bit                                               |
|                             | Accuracy                    | ±(0.015%+0.1% FS) ±(0.015%+0.1% FS)                         |
| CW Mode                     | Range                       | 150W 300W                                                   |
|                             | Resolution                  | 0.01W 0.01W                                                 |
|                             | Accuracy                    | ±(0.1%+0.1% FS) ±(0.1%+0.1% FS)                             |
| Slope                       | Range                       | 0-3A 0-30A 0-3A 0-30A                                       |
|                             | Rising                      | 0.0001-0.15A/us 0.001-1.5A/us 0.0001-0.15A/us 0.001-1.5A/us |
|                             | Falling                     | 0.0001-0.15A/us 0.001-1.5A/us 0.0001-0.15A/us 0.001-1.5A/us |
| Voltage measurement         | Range                       | 0-18V 0-120V 0-18V 0-120V                                   |
|                             | Resolution                  | 0.1mV 1mV 0.1mV 1mV                                         |
|                             | Accuracy                    | ±(0.025%+0.025% FS) ±(0.025%+0.025% FS)                     |
| Current measurement         | Range                       | 0-3A 0-30A 0-3A 0-30A                                       |
|                             | Resolution                  | 0.1mA 1mA 0.1mA 1mA                                         |
|                             | Accuracy                    | ±(0.05%+0.05% FS) ±(0.05%+0.05% FS)                         |
| Power measurement           | Range                       | 150W 300W                                                   |
|                             | Resolution                  | 0.01W 0.01W                                                 |
|                             | Accuracy                    | ±(0.1%+0.1% FS) ±(0.1%+0.1% FS)                             |
| Over power protection       |                             | 160W 320W                                                   |
| Over current protection     |                             | 3.3A/33A 3.3A/33A                                           |
| Over voltage protection     |                             | 125V 125V                                                   |
| Over temperature protection |                             | 85°C 85°C                                                   |
| Input impedance             |                             | 150KΩ 150KΩ                                                 |
| Interfaces                  |                             | USB, RS232 & LAN                                            |
| Weight and Dimention(W*D*H) |                             | 5.35KGS / 214mm*354mm*88.5mm                                |

Note: Specifications are subject to change without notice.

Remarks: After the keyboard is lock, there are only ON/OFF and Shift which are valid. And during the dynamic pulse mode and dynamic toggling mode, the button 3 is valid. And there is no memory function.

15. Baud Rate Setting Function

Note: Manually modify the baud rate.  
Operation Description: taking changing the baud rate 57600 into 115200 as example.

| Procedures | Operation Description                                                             | Device Displays                    |
|------------|-----------------------------------------------------------------------------------|------------------------------------|
| 1          | Press “ <b>Shift</b> + <b>0</b> ” to enter the setting mode of the baud rate.     | 0.0000V 0.0000A<br>Bsp-0 57600G    |
| 2          | Rotate the knob or press the up and down buttons to adjust the display as 115200. | 0.0000V 0.0000A<br>Bsp-1 15200G    |
| 3          | Press “ <b>ENTER</b> ” to confirm.                                                | Returning to the steady state mode |

Remarks: please modify when there are no communications.  
Here there is memory function.

16. Buzzer ON/OFF Function

Note: press Shift+① to switch the buzzer on and off.  
Remarks: there are memory function here.

17. Communication with PC

Please refer to the communication protocol.

Remarks: When the keyboard is not locked, press Esc at any time to return to the steady state mode. From one mode switched to another mode, press Esc to enter the steady state mode and then switch the modes.

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| Procedures | Operation Description                                                                                                                                         | Device Displays                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1          | Press “CALL + 5” mode of recalling the setting sequence.                                                                                                      | 0.0000V 0.0000A<br>CALL P-001G  |
| 2          | “ENTER” 1 by pressing the knob or buttons to recall.                                                                                                          | 0.0000V 0.0000A<br>CALL P-001G  |
| 3          | Press “ON/OFF” to start or repeat. When it meets one of the conditions, the load output will be closed automatically. And then press “ON/OFF” again to start. | 0.0000V 0.0000A<br>PASS 12.000W |

Remarks: press ON/OFF to start and then the LED display on the left bottom is whether it passes or not; and that on the right bottom is the set power value at present.

**12. External Trigger Function**

Note: Press Shift+CR to switch the external trigger function. Through the remote control, the steady state mode can be opened; the test mode of the dynamic state, the sequence operation mode and the battery test mode can be triggered.

Remarks: when LED GRAP on the display lights on, it can be triggered externally. And here there is the memory function, that is, the trigger will be saved.

**13. Remote Compensation Function**

Note: Press Shift+CW to enter the switch of the remote compensation function.

Remarks: When there is the remote compensation, the LED COMM will light on. And there is no memory function.

**14. Keyboard Lock Function**

Note: Press Shift+⊙ to make the keyboard lock.

|    |                                                                                                                                        |                                  |                  |
|----|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------|
| 7  | Enter the step power 1W by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.          | 0.0000V<br>0V                    | 0.0000A<br>P0107 |
| 8  | Enter the step reduction time 1s by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings. | 0.0000V<br>0V                    | 0.0000A<br>P0108 |
| 9  | Enter the ending power 10W by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.       | 0.0000V<br>0V                    | 0.0000A<br>P0109 |
| 10 | Enter the OPP voltage 8V by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.         | 0.0000V<br>0V                    | 0.0000A<br>P0110 |
| 11 | Enter the maximum power 15W by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.      | 0.0000V<br>0V                    | 0.0000A<br>P0110 |
| 12 | Enter the minimum power 10W by pressing the knob or buttons and then press “ <b>ENTER</b> ” to finish the settings.                    | Return to the steady state mode. |                  |

Remarks: During the operation, once press Esc and it will exit the mode. If there are some wrong entering, you can rotate the knob. When the complete value appears, enter again.

11.2 The Recall Function of OPP Test

Note: recall one group of storage and use it.  
Operation Description: recall the first group of storage and use it.

**SAFETY INSTRUCTION**

Safety Symbols

These safety symbols may appear in this manual or on the series.



WARNING



DANGER High Voltage



Earth(ground)Terminal

**Safety Guidelines**

- Do not block or obstruct the cooling fan vent opening
- Avoid severe impacts or rough handling that leads to damage.
- Do not discharge static electricity.
- Do not disassemble unless you are qualified as service personnel.

**AC INPUT**



- AC Input Voltage: 110V / 120V / 220V / 230V , 50 / 60 Hz
- Connect the protective grounding conductor of the AC power cord to an earth ground, to avoid electrical shock.

**Operation**

**Environment**

- Location: Indoor, no direct sunlight, dust free, no conductive pollution
- Relative Humidity: < 80%
- Altitude: < 2000m
- Temperature: 0-40°C

**Storage**

**environment**

- Location: Indoor
- Relative Humidity: < 70%
- Temperature: -10-70°C

**FUSE**



| Model    | 110V/120V           | 220V/230V          |
|----------|---------------------|--------------------|
| MP710258 | T1.25A/250V(20X5mm) | T0.5A/250V(20x5mm) |
| MP710259 | T1.25A/250V(20X5mm) | T0.5A/250V(20x5mm) |

- To ensure fire protection, replace the fuse only with the specified type and rating.
- Disconnect the power cord before fuse replacement.
- Make sure the cause of fuse blowout is confirmed before fuse replacement.

Note: It can set at most 10 groups of test parameters.  
Operation description: taking it as example that the setting is saved in group 1, VON voltage is 10V, time delay of VON voltage is 5s, the current range is 3A, the beginning power is 20W, the step reduction value is 1W, the reduction time is 1s, the ending power is 10W, OPP voltage is 8V, the maximum power is 15W and the minimum power is 10W.

| Procedures | Operation Description                                                                                                               | Device Displays                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1          | Press “ <b>shift</b> + <b>5</b> ” and enter the storage mode of OPP settings.                                                       | 0.0000V    0.0000A<br>0    P0001  |
| 2          | Enter 1 by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the storage mode of the first group.               | 0.0000V    0.0000A<br>0V    P0102 |
| 3          | Enter VON voltage 10V by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.         | 0.0000V    0.0000A<br>0V    P0103 |
| 4          | Enter the VON time delay 5s by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.   | 0.0000V    0.0000A<br>0V    P0104 |
| 5          | Enter the current range 3A by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.    | 0.0000V    0.0000A<br>0V    P0105 |
| 6          | Enter the beginning power 20W by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings. | 0.0000V    0.0000A<br>0V    P0106 |

10.2 The Recall Function of OCP Test

Note: recall one group of storage and use it.  
Operation Description: recall the first group of storage and use it.

| Procedures | Operation Description                                                                                                                                                           | Device Displays                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1          | Press “ <b>CALL</b> + <b>9</b> ” and enter the mode of recalling the setting sequence.                                                                                          | 0.0000V    0.0000A<br>CALL       C-001G |
| 2          | “ <b>ENTER</b> ” 1 by pressing the knob or buttons to recall.                                                                                                                   | 0.0000V    0.0000A<br>CALL       C-001G |
| 3          | Press “ <b>ON/OFF</b> ” to start or repeat. When it meets one of the conditions, the load output will be closed automatically. And then press “ <b>ON/OFF</b> ” again to start. | 0.0000V    0.0000A<br>PASS       2.0000 |

Remarks: press ON/OFF to start and then the LED display on the left bottomis whether it passes or not; and that on the right bottom is the set current value at present.

11. OPP Test Function

Note: when the voltage reaches the VON value, it will delay for some time, and the power outputs; at intervals, the step value will decrease progressively until it reaches the cutoff power, or the voltage is higher than that set by OPP, and then the output ends. After that, if the voltage is higher than the OPP voltage and meanwhile the power value is between the maximum and the minimum set powers, it passes; otherwise, it fails.

11.1 The Setting Function of OPP Test

Note: It can set at most 10 groups of test parameters.

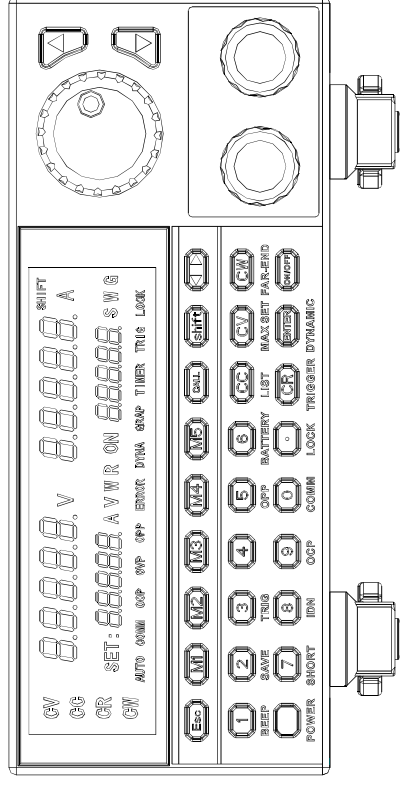
Introduction

MP710258 series (150W~300W), single-channel programmable DC electronic loads, are designed for middle & high-end applications. They can be offered as multiple solutions according to customer's need. The online voltage measurements and adjustments or simulating short circuit test using the simple keypad on the front panel, can be used by the end users. MP710258 series DC loads are a versatile instrument for static and dynamic testing of power supplies, batteries, DC - DC converters, and battery chargers, which provides users the best testing solution.

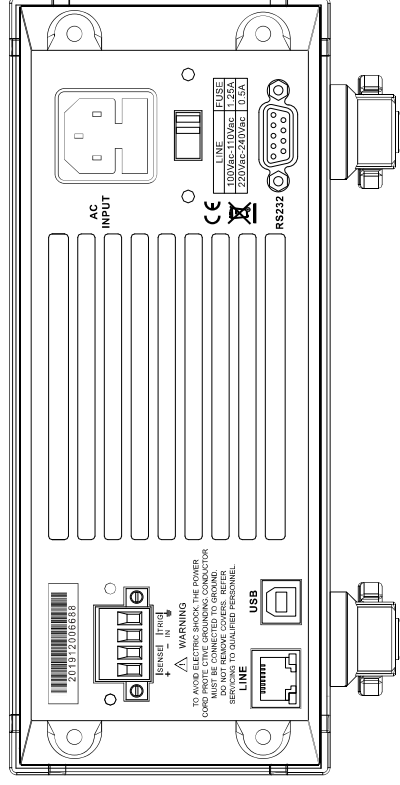
Main Features

- Highlight LED display
- 5-digit display, and accurate outputs
- Resolution of voltage and current: 0.1 mV / 0.1 mA
- Over-voltage Protection
- 100 groups of memories for fast recall
- Four working modes: CV/CC/CR/CP
- Remote Compensation Function
- Battery Test Function
- Keyboard Lock
- Power off memory function
- Short-circuit function
- External Trigger Function
- Setting Function of Baud Rate

# Front Panel Overview



## Rear Panel Overview



|    |                                                                                                                                      |                                  |
|----|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 7  | Enter the step current 0.1A by pressing the knob or buttons and then press " <b>ENTER</b> " to enter the next parameter settings.    | 0.0000V 0.0000A<br>0V C0107      |
| 8  | Enter the reduction time 1s by pressing the knob or buttons and then press " <b>ENTER</b> " to enter the next parameter settings.    | 0.0000V 0.0000A<br>0V C0108      |
| 9  | Enter the ending current 1A by pressing the knob or buttons and then press " <b>ENTER</b> " to enter the next parameter settings.    | 0.0000V 0.0000A<br>0V C0109      |
| 10 | Enter the OCP voltage 8V by pressing the knob or buttons and then press " <b>ENTER</b> " to enter the next parameter settings.       | 0.0000V 0.0000A<br>0V C01010     |
| 11 | Enter the maximum current 1.9A by pressing the knob or buttons and then press " <b>ENTER</b> " to enter the next parameter settings. | 0.0000V 0.0000A<br>0V C0111      |
| 12 | Enter the minimum current 1.1A by pressing the knob or buttons and then press " <b>ENTER</b> " to finish the settings.               | Return to the steady state mode. |

Remarks: During the operation, once press Esc and it will exit the mode. If there are some wrong entering, you can rotate the knob. When the complete value appears, enter again.

Operation description: taking it as example that the setting is saved in group 1, VON voltage is 10V, time delay of VON voltage is 5s, the current range is 3A, the beginning current is 2A, the step reduction value is 0.1A, the reduction time is 1s, the ending current is 1A, OCP voltage is 8V, the maximum current is 1.9A and the minimum current is 1.1A.

| Procedures | Operation Description                                                                                                                | Device Displays             |
|------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 1          | Press “ <b>shift</b> + <b>9</b> ” and then enter the storage mode of OCP settings.                                                   | 0.0000V 0.0000A<br>0 C0001  |
| 2          | Enter 1 by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the storage mode of the first group.                | 0.0000V 0.0000A<br>0V C0102 |
| 3          | Enter VON voltage 10V by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.          | 0.0000V 0.0000A<br>0V C0103 |
| 4          | Enter the VON time delay 5s by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.    | 0.0000V 0.0000A<br>0V C0104 |
| 5          | Enter the current range 3A by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.     | 0.0000V 0.0000A<br>0V C0105 |
| 6          | Enter the beginning current 2A by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings. | 0.0000V 0.0000A<br>0V C0106 |

Remarks: press ON/OFF to start and then the LED display on the right bottom is the count of triggers.

## Operation

### 1. Keyboard Operation

- 1.Esc: Cancel key which can return to the initial mode.
2. M1-M5: quickly recalling the M1-M5 stored value of the steady state.
3. Recall: Recall Key, such as, recalling the steady storage with Call+2.
4. Shift: Setting or Storage Function Keys, such as storing the stationary content with Shift+2.
5. **↔** — Left-Rotating Key, moving to where needs to be adjusted and will flash.
6. 0-9 Entering numbers.
7. CC-Constant Current Mode Button.
8. CC-Constant Voltage Mode Button.
9. CW-Constant Power Mode Button.
10. CR-Constant Resistance Mode Button.
11. Confirmation Button.
12. On/Off: Turning On/Off the load output.
13. **▲** — Adjusting the number or selecting add.
14. **▼** — Adjusting the number or selecting subtract.
15. Knob: Adjusting the number or selecting it.

### 2. Setting the maximum value of load

Description: Setting low voltage (18V), low current (3A) and below them the alwracy will be improved.  
Method of operation: Taking the max current 3A, max voltage 18V and max power 100W for example, the max resistance is 1000ohms.

| Procedures | Operation Description                                         | Device Displays                      |
|------------|---------------------------------------------------------------|--------------------------------------|
| 1          | Press “ <b>shift</b> + <b>CV</b> ” to operate                 | 0.0000V 0.000A<br>SET:30.000A 00000G |
| 2          | Setting the max current value and then press “ <b>ENTER</b> ” | 0.0000V 0.000A<br>SET:3.0000A 00000G |

|   |                                                                  |                                      |                  |
|---|------------------------------------------------------------------|--------------------------------------|------------------|
| 3 | Press “ <b>CV</b> ” to operate                                   | 0.0000V<br>SET:120.00V               | 0.000A<br>00000G |
| 4 | Setting the max voltage value and then press “ <b>ENTER</b> ”    | 0.0000V<br>SET:18.000V               | 0.000A<br>00000G |
| 5 | Press “ <b>CV</b> ” to operate                                   | 0.0000V<br>SET:300.00W               | 0.000A<br>00000G |
| 6 | Setting the max power value and then press “ <b>ENTER</b> ”      | 0.0000V<br>SET:100.00W               | 0.000A<br>00000G |
| 7 | Press “ <b>CV</b> ” to operate                                   | 0.0000V<br>SET:7500.0R               | 0.000A<br>00000G |
| 8 | Setting the max resistance value and then press “ <b>ENTER</b> ” | 0.0000V<br>SET:1000.0R               | 0.000A<br>00000G |
| 9 | After the setting is finished, press “ <b>Esc</b> ” to exit.     | The steady-mode status is displayed. |                  |

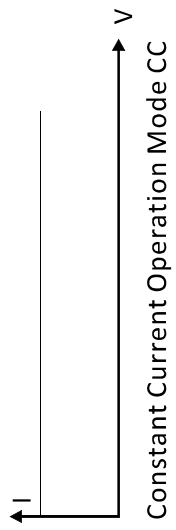
3. The Operation Function of Steady State

Note: The electronic load can work in the following 4 steady-state modes.  
3.1 Constant Current Operation Mode CC

Note: In constant current mode, the load makes the tested equipment on the set voltage no matter how the input voltage changes.  
Operation Method:

- 1. Press the CC button on the keyboard to enter constant current operation mode.
- 2. Setting the desired constant current through the keyboard.
- 3. Turn on the ON/OFF button to start the electronic load.

Load Current



9.2 The Recall Function of the Battery Test

Note: Recall one group of the storage and use it.  
Operation Description: Recall the first group of storage and use it.

| Procedures | Operation Description                                                                                                                                                 | Device Displays                  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 1          | Press “ <b>CALL</b> + <b>6</b> ” and enter the recall mode of the setting sequence.                                                                                   | 0.0000V 0.0000A<br>CALL b-001G   |
| 2          | “ <b>ENTER</b> ” 1 by pressing the knob or buttons to recall.                                                                                                         | 0.0000V 0.0000A<br>CALL b-001G   |
| 3          | Press “ <b>ON/OFF</b> ” to start or repeat. When meeting one of the conditions, the output will close automatically. And then press “ <b>ON/OFF</b> ” to start again. | 0.0000V 0.0000A<br>1.0000 0.5000 |

Remarks: Press ON/OFF to start and then the LED display on the left bottom is the running time and the LED display on the right bottom is the consumption of capacity.

10. OCP Test Function

Note: when the voltage reaches the VON value, it will delay for some time, and the current outputs; at intervals, the step value will decrease progressively until it reaches the cutoff current, or the voltage is higher than that set by OCP, and then the output ends. After that, if the voltage is higher than the OCP voltage and meanwhile the current value is between the maximum and the minimum set currents, it passes; otherwise, it fails.

10.1 Function of OCP Test Settings

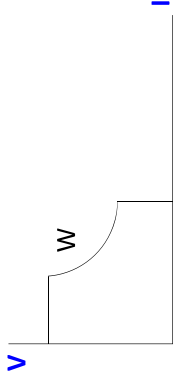
Note: It can set at most 10 groups of test parameters.



3.4 Constant Power Operation Mode CW

Note: In constant power mode, the load makes the tested equipment on the set resistance no matter how the input voltage and current change.  
Operation Method:

- 1. Press the CW button on the keyboard to enter constant power operation mode.
- 2. Setting the desired constant power through the keyboard.
- 3. Turn on the ON/OFF button to start the electronic load.



Constant Power Operation Mode CW

4. The Storage Function of Steady State and Recalling

Note: The load can save and recall 100 sets of stationary setting values. And to set the values, the number keys, knob and up & down buttons all can be used.

4.1 Storage function

Operation Method: Taking 10.000V stored to 99 as example.

| Procedures | Operation Description                                                  | Device Displays                   |
|------------|------------------------------------------------------------------------|-----------------------------------|
| 1          | Setting the steady state value to be stored(10.000V)                   | 0.0000V 0.0000A<br>10.000V 00000G |
| 2          | Press “ <b>shift</b> + <b>2</b> ” to enter the memory function.        | 0.0000V 0.0000A<br>10.000V 00001G |
| 3          | Using the button input or the knob to 99 and then press ENTER to store | 0.0000V 0.0000A<br>10.000V 00001G |

|    |                                 |
|----|---------------------------------|
| 15 | Press Esc to exit the settings. |
|----|---------------------------------|

Remarks:During the operation,one press Esc and it will exit the mode. If there are some wrong entering, you can rotate the knob.When the complete value appears, enter again.

8.2 Sequence Recall Function

Note: recall one group of storage and use it.  
Operation Description: recall the first group of storage and use it.

| Procedures | Operation Description                                                                                                                                                       | Device Displays                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1          | Press “ <b>CALL</b> + <b>CC</b> ” and enter the mode of recalling the setting sequence.                                                                                     | 0.0000V 0.0000A<br>CALL L-001G  |
| 2          | “ <b>ENTER</b> ” 1 by pressing the knob or buttons to recall.                                                                                                               | 0.0000V 0.0000A<br>CALL L-001G  |
| 3          | Press “ <b>ON/OFF</b> ” to start or pause. When the repetition times finish, the load output will be closed automatically. And then press “ <b>ON/OFF</b> ” again to start. | 0.0000V 0.0000A<br>00005 L-001G |

Remarks: press ON/OFF to start and then the LED display on the right bottom is the times of repetition.

9.Battery Test Function

9.1 The Setting Function of the Battery Test

Note: It can at most set 10 groups of battery test parameters. According to the set current, voltage, capacity and time, it tests the battery. And the test will be turned off automatically once it meets any one of the conditions.

Operation Description: taking it as example that the setting is saved in group 1, the current range is 10A, the discharge current is 1A, the discharging end-off voltage is 2A, the discharging end-off capacity is 0.5AH and the discharging time is 200s.

|    |                                                                                                                                           |               |                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|
| 6  | Enter the change rate 0.001A/us by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.     | 0.0000V<br>0  | 0.0000A<br>L1006 G |
| 7  | Enter the time 1s by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.                   | 0.0000V<br>0A | 0.0000A<br>L1007 G |
| 8  | Enter the second dynamic current 2A by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings. | 0.0000V<br>0A | 0.0000A<br>L1008G  |
| 9  | Enter the change rate 0.002A/us by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.     | 0.0000V<br>0  | 0.0000A<br>L1009G  |
| 10 | Enter the time 2s by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.                   | 0.0000V<br>0A | 0.0000A<br>L1010 G |
| 11 | Enter the third dynamic current 3A by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.  | 0.0000V<br>0A | 0.0000A<br>L1011G  |
| 12 | Enter the change rate 0.003A/us by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.     | 0.0000V<br>0  | 0.0000A<br>L1012G  |
| 13 | Enter the time 3s by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.                   | 0.0000V<br>0  | 0.0000A<br>L1013G  |
| 14 | Enter the times 5 by pressing the knob or buttons and then press “ <b>ENTER</b> ” to finish the settings and storage.                     | 0.0000V<br>5  | 0.0000A<br>L1013G  |

## 4.2 Recall Function

Operation method: taking recalling the stored steady-state value in 99 As example.

| Procedures | Operation Description                                                         | Device Displays                   |
|------------|-------------------------------------------------------------------------------|-----------------------------------|
| 1          | Press “ <b>CALL</b> + <b>2</b> ” to enter the recall function                 | 0.0000V 0.0000A<br>10.000V 00000G |
| 2          | Using the button input or the knob to 99 and then press <b>ENTER</b> to store | 0.0000V 0.0000A<br>10.000V 00099G |

## 5. M1-M5 Fast Recall and Storage Function

### 5.1 M1-M5 Fast Recall Function

Note: By pressing M1-M5, the steady state content stored in M1-M5 can be quickly recalled.

### 5.2 M1-M5 Fast Storage Function

Note: You can change the content of M1-M5 by selecting M1-M5 button; and then press the currently selected M1-M5 button to fast save the value to the currently selected M key.

For example, the previous content of M1 is CV+15.001V and now needing to modify as CV+14.000V, you can press M1 to recall the content and then modify to 14.000V through rotating the knob and then pressing the number keys to select the value you want. After that, press M1 again to save it in M1.

## 6. Short Circuit Function

Note: the load will make the tested equipment output the max current. Operation Method:

| Procedures | Operation Description                                                     | Device Displays                   |
|------------|---------------------------------------------------------------------------|-----------------------------------|
| 1          | Press “ <b>shift</b> + <b>7</b> ” to enter the short circuit testing mode | 0.0000V 0.0000A<br>10.000V 00000G |

## 7. Dynamic Test Function

Note: There are 6 setting functions in this mode: Dynamic CV, Dynamic CC, Dynamic CR, Dynamic CW, Dynamic Pulse and Dynamic Flip. And this mode has no storage function, so it can be only set first and then run.

### 7.1 Dynamic CV, Dynamic CR, & Dynamic CW

Description: used for the different duty cycle output of 2 different voltages at a certain frequency.

Operation Method: Taking the first voltage 1V, the second voltage 2V, the cycle frequency 1HZ and the duty cycle 40% as example; as for Dynamic CR and Dynamic CW, just need to change the setting voltage into resistance or power. And Dynamic CR selects mode 3 while Dynamic CW selects mode 4.

| Procedures | Operation Description                                                                                                                   | Device Displays                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1          | Press the buttons “ <b>shift</b> + <b>ENTER</b> ” to enter the setting mode of dynamic selection                                        | 0.0000V 0.0000A<br>0.0000 00001 G |
| 2          | “ <b>ENTER</b> ” 1 through the knob or buttons and then press <b>ENTER</b> to enter the dynamic CV setting mode                         | 0.0000V 0.0000A<br>0V 1-001 G     |
| 3          | Using buttons or the knob to enter the first voltage 1V and then press the button “ <b>ENTER</b> ” to enter the next parameter setting  | 0.0000V 0.0000A<br>0V 1-002 G     |
| 4          | Using buttons or the knob to enter the second voltage 2V and then press the button “ <b>ENTER</b> ” to enter the next parameter setting | 0.0000V 0.0000A<br>0 1-003 G      |
| 5          | Using buttons or the knob to enter the frequency 1HZ and then press the button “ <b>ENTER</b> ” to enter the next parameter setting     | 0.0000V 0.0000A<br>0 1-004 G      |

Remarks: press ON/OFF to start and then the LED display on the right bottom is the count of triggers.

## 8.Sequential Operation Function

### 8.1 Sequential Setting Function

Note: It can save 7 groups at most and every set it can at most set 84 dynamic currents. And the set current can be toggled in sequence.

Operation description: taking it as example that the setting is saved in group 1, the max current is 4A, there are 3 dynamic currents, the first dynamic current is 1A, the change rate is 0.001A/us, the time is 1s, the second dynamic current is 2A, the change rate is 0.002A/us, the time is 2s, the first dynamic current is 3A, the change rate is 0.003A/us, the time is 3S and the repetition times is 5 times.

| Procedures | Operation Description                                                                                                                     | Device Displays              |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1          | Press the button “ <b>shift</b> + <b>CC</b> ” and enter the storage mode of the sequence setting.                                         | 0.0000V 0.0000A<br>0 L1001 G |
| 2          | Enter 1 through the knob or buttons and then press “ <b>ENTER</b> ” to enter the first group of setting mode.                             | 0.0000V 0.0000A<br>0 L1002 G |
| 3          | Using buttons or the knob to enter the max current 4A and then press the button “ <b>ENTER</b> ” to enter the next parameter setting.     | 0.0000V 0.0000A<br>0 L1003 G |
| 4          | Enter 3 currents of dynamic changes by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings. | 0.0000V 0.0000A<br>0 L1004 G |
| 5          | Enter the first dynamic current 1A by pressing the knob or buttons and then press “ <b>ENTER</b> ” to enter the next parameter settings.  | 0.0000V 0.0000A<br>0 L1005 G |

#### 7.4 Dynamic Toggle

Note: Every time when receiving a trigger signal, the load will toggle between the first and second setting current.

Operation method: taking it as example that the change slope of the first current is 0.001A/Us, that of the second current is 0.002A/Us, the first current is 1A, and the second current is 2A.

| Procedures | Operation Description                                                                                                                                       | Device Displays                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 1          | Press the buttons “ <b>shift</b> + <b>ENTER</b> ” to enter the setting mode of dynamic selection.                                                           | 0.0000V 0.0000A<br>0.0000 00001 G    |
| 2          | Enter 6 through the knob or buttons and then press “ <b>ENTER</b> ” to enter the setting mode of the dynamic pulse.                                         | 0.0000V 0.0000A<br>0A 6-001 G        |
| 3          | Using buttons or the knob to enter the first current change rate 0.001A/us and then press the button “ <b>ENTER</b> ” to enter the next parameter setting.  | 0.0000V 0.0000A<br>0A 6-002 G        |
| 4          | Using buttons or the knob to enter the second current change rate 0.002A/us and then press the button “ <b>ENTER</b> ” to enter the next parameter setting. | 0.0000V 0.0000A<br>0A 6-003 G        |
| 5          | Using buttons or the knob to enter the first current 1A and then press the button “ <b>ENTER</b> ” to enter the next parameter setting.                     | 0.0000V 0.0000A<br>0A 6-004 G        |
| 6          | Using buttons or the knob to enter the second current 2A and then press the button “ <b>ENTER</b> ” to finish the settings.                                 | 0.0000V 0.0000A<br>0 6-000 G<br>TRIG |
| 7          | Press the button “ <b>ON/OFF</b> ” to start or pause and then press the button 3 to trigger and toggle once.                                                | 0.0000V 0.0000A<br>0 5-000 G<br>TRIG |

|   |                                                                                                              |                                          |
|---|--------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 6 | Using buttons or the knob to enter the duty cycle 40% and then press “ <b>ENTER</b> ” to finish the settings | 0.0000V 0.0000A<br>00000 1-000 G<br>TRIG |
| 7 | Press the button “ <b>ON/OFF</b> ” to start or pause                                                         |                                          |

Remarks: After pressing ON/OFF to start, the LED display on the right bottom is the count running once.

#### 7.2 Dynamic CC

Note: used for the different duty cycle output of 2 different currents at a certain frequency.

Operation Method: taking it as example that the change slope of the first current is 0.001A/Us, that of the second current is 0.002A/Us, the first current is 1A, the second current is 2A, the cycle frequency is 1HZ and the duty cycle is 40%.

| Procedures | Operation Description                                                                                                                                      | Device Displays                   |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1          | Press the buttons “ <b>shift</b> + <b>ENTER</b> ” to enter the setting mode of dynamic selection                                                           | 0.0000V 0.0000A<br>0.0000 00001 G |
| 2          | “ <b>ENTER</b> ” 2 through the knob or buttons and then press <b>ENTER</b> to enter the dynamic CC setting mode                                            | 0.0000V 0.0000A<br>0A 2-001 G     |
| 3          | Using buttons or the knob to enter the first current change rate 0.001A/us and then press the button “ <b>ENTER</b> ” to enter the next parameter setting  | 0.0000V 0.0000A<br>0A 2-002 G     |
| 4          | Using buttons or the knob to enter the second current change rate 0.002A/us and then press the button “ <b>ENTER</b> ” to enter the next parameter setting | 0.0000V 0.0000A<br>0A 2-003 G     |

|   |                                                                                                                                         |                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 5 | Using buttons or the knob to enter the first current 1A and then press the button “ <b>ENTER</b> ” to enter the next parameter setting  | 0.0000V<br>0A<br>2-004 G            |
| 6 | Using buttons or the knob to enter the second current 2A and then press the button “ <b>ENTER</b> ” to enter the next parameter setting | 0.0000V<br>0A<br>2-005 G            |
| 7 | Using buttons or the knob to enter the cycle 1HZ and then press the button “ <b>ENTER</b> ” to enter the next parameter setting         | 0.0000V<br>0A<br>2-006 G            |
| 8 | Using buttons or the knob to enter the duty cycle 40% and then press the button “ <b>ENTER</b> ” to finish the settings                 | 0.0000V<br>00000<br>2-000 G<br>TRIG |
| 9 | Press the button “ <b>ON/OFF</b> ” to start or pause                                                                                    | 0.0000V<br>00001<br>2-000 G<br>TRIG |

Remarks: After pressing ON/OFF to start, the LED display on the right bottom is the count running once.

### 7.3 Dynamic Pulse

Note: at the beginning, it is the first setting current. And every time when receiving a trigger signal, the load will switch to the second setting current. After maintaining the setting time, it will switch to the first current. Operation method: taking it as example that the change slope of the first current is 0.001A/Us, that of the second current is 0.002A/Us, the first current is 1A, the second current is 2A, and the second current maintenance time is set as 1s.

| Procedures | Operation Description                                                                                                                                       | Device Displays                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1          | Press the buttons “ <b>shift</b> + <b>ENTER</b> ” to enter the setting mode of dynamic selection.                                                           | 0.0000V<br>0.0000<br>00001 G    |
| 2          | Enter 5 through the knob or buttons and then press “ <b>ENTER</b> ” to enter the setting mode of the dynamic pulse.                                         | 0.0000V<br>0A<br>5-001 G        |
| 3          | Using buttons or the knob to enter the first current change rate 0.001A/us and then press the button “ <b>ENTER</b> ” to enter the next parameter setting.  | 0.0000V<br>0A<br>5-002 G        |
| 4          | Using buttons or the knob to enter the second current change rate 0.002A/us and then press the button “ <b>ENTER</b> ” to enter the next parameter setting. | 0.0000V<br>0A<br>5-003 G        |
| 5          | Using buttons or the knob to enter the first current 1A and then press the button “ <b>ENTER</b> ” to enter the next parameter setting.                     | 0.0000V<br>0A<br>5-004 G        |
| 6          | Using buttons or the knob to enter the second current 2A and then press the button “ <b>ENTER</b> ” to enter the next parameter setting.                    | 0.0000V<br>0<br>5-005 G         |
| 7          | Using buttons or the knob to enter the pulse width and then press “ <b>ENTER</b> ” to finish the settings.                                                  | 0.0000V<br>0<br>5-000 G<br>TRIG |
| 8          | Press the button “ <b>ON/OFF</b> ” to start or pause and then press the button 3 to trigger once.                                                           | 0.0000V<br>0<br>5-000 G<br>TRIG |

Remarks: press ON/OFF to start and then the LED display on the right bottom is the count of triggers.